

Experiment No. 2

Aim: To convert the E-R diagram of an Indian E-Commerce platform into a relational schema, create MySQL tables with PRIMARY KEY, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints, insert sample data and demonstrate referential integrity violations.

Relational Schema

Table	Attributes
Student	Student_Id (PK), Student_Name, Email (Unique), Dept_Id (FK)
Department	Dept_Id (PK), Dept_Name
Course	Course_Id (PK), Course_Name, Credits
Enrollment	Enroll_Id (PK), Student_Id (FK), Course_Id (FK)

Source Code (MySQL 8)

```
CREATE DATABASE CollegeDB;
USE CollegeDB;

CREATE TABLE Department (
    Dept_Id INT PRIMARY KEY,
    Dept_Name VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE Student (
    Student_Id INT PRIMARY KEY,
    Student_Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE,
    Dept_Id INT,
    FOREIGN KEY (Dept_Id)
        REFERENCES Department(Dept_Id)
        ON DELETE SET NULL
);

CREATE TABLE Course (
    Course_Id INT PRIMARY KEY,
    Course_Name VARCHAR(50) NOT NULL,
    Credits INT NOT NULL
);

CREATE TABLE Enrollment (
    Enroll_Id INT PRIMARY KEY,
    Student_Id INT,
    Course_Id INT,
    FOREIGN KEY (Student_Id)
        REFERENCES Student(Student_Id)
        ON DELETE CASCADE,
    FOREIGN KEY (Course_Id)
        REFERENCES Course(Course_Id)
        ON DELETE CASCADE
);
```

Inserting Sample Data

```
INSERT INTO Department VALUES (1, 'CSE'), (2, 'AIML');

INSERT INTO Student VALUES
```

```
(101, 'Rahul', 'rahul@gmail.com', 1),  
(102, 'Vansh', 'vansh@gmail.com', 2);
```

```
INSERT INTO Course VALUES
```

```
(201, 'DBMS', 4),  
(202, 'Operating System', 4);
```

```
INSERT INTO Enrollment VALUES (1, 101, 201), (2, 102, 202);
```

Output

Department

Dept_Id	Dept_Name
1	CSE
2	AIML

Student

Student_Id	Name	Email	Dept_Id
101	Rahul	rahul@gmail.com	1
102	Vansh	vansh@gmail.com	2

Course

Course_Id	Name	Credits
201	DBMS	4
202	Operating System	4

Enrollment

Enroll_Id	Student_Id	Course_Id
1	101	201
2	102	202

Referential Integrity Violation (Demonstration)

```
-- Student with a non-existent department
INSERT INTO Student VALUES (103, 'Aman', 'aman@gmail.com', 9);
-- ERROR 1452: Cannot add or update a child row: a foreign key constraint fails

-- Enrollment for a non-existent course
INSERT INTO Enrollment VALUES (3, 101, 999);
-- ERROR 1452: Cannot add or update a child row: a foreign key constraint fails

-- Duplicate email (UNIQUE violation)
INSERT INTO Student VALUES (104, 'Neha', 'rahul@gmail.com', 1);
-- ERROR 1062: Duplicate entry 'rahul@gmail.com' for key 'Student.Email'

-- ON DELETE SET NULL: deleting a department
DELETE FROM Department WHERE Dept_Id = 1;
-- Rahul's Dept_Id becomes NULL

-- ON DELETE CASCADE: deleting a student
DELETE FROM Student WHERE Student_Id = 102;
-- Vansh's row in Enrollment is deleted automatically
```